

AI Skepticism & User Trust — Project Proposal — Aditi Shelke

MISM 6412 — Data Mining / Machine Learning for Business

Aditi Shelke & Muskaan Agarwal | Project Proposal

Business Question

- Explore how users interact with AI-generated responses and how different factors influence trust, skepticism, and verification behavior — specifically how AI confidence levels compare with actual answer accuracy and whether this relationship affects user trust.
- Analyze how individual user differences (education level, age bracket, digital literacy) shape trust behavior, skepticism categories, and fact-checking patterns.
- Examine temporal trends by incorporating year and quarter variables to understand whether trust dynamics evolve over time as users become more familiar with AI technologies.
- Determine whether AI systems are fostering a net positive trust effect across different demographic groups and what improvements can enhance responsible, equitable AI adoption.

Data Sources

- Sourced from Kaggle: Human Trust Levels in AI Systems (kaggle.com/datasets/shaistashahid/human-trust-levels-in-ai-systems). Minimal cleaning required as data was pre-structured.
- Time-based variables (year, quarter) were added to enable trend analysis of trust scores, skepticism, and AI calibration patterns over time.
- Key variables: AI model name, AI confidence %, answer accuracy %, trust score, calibration validity, skepticism category, cited sources, disclaimers, hedging language, response length, detail level.
- User-level variables: age bracket, education level, digital literacy score, AI familiarity, subject-matter expertise, decision importance, fact-check behavior, verification duration.

Data Quality Considerations

- Missing values present in fields such as urgency level and belief alignment status — handled carefully to avoid distorted results.
- Dataset sourced from Kaggle may be simulated or aggregated rather than from controlled experiments, limiting external validity.

- Self-reported variables (trust score, digital literacy, AI familiarity) may introduce response bias.
- Year and quarter variables were artificially added — temporal patterns should be interpreted cautiously as they do not represent naturally occurring time-series data.

Analytical Methods

- Data wrangling: cleaning, missing value handling, encoding categorical variables (education level, age bracket, skepticism category).
- Exploratory Data Analysis (EDA): distributions, correlations, initial pattern identification.
- Trend analysis using year/quarter variables to track how trust scores, skepticism, and fact-checking behavior evolve over time.
- Multiple regression to assess how AI confidence, answer accuracy, digital literacy, education, and age influence trust scores.
- Logistic regression for categorical outcomes: fact-checking behavior and skepticism category classification.
- Hypothesis testing and model validation (pROC, cross-validation) to evaluate statistical significance and robustness.

Team Contributions

- Aditi Shelke: Data cleaning, EDA, visualization of trends, regression modeling, hypothesis testing using R/Python, model validation.
- Muskaan Agarwal: Data integration, interpretation of results, report writing, presentation preparation.

All analyses, report writing, and presentation preparation will be shared equally across both team members.